CAMPUS PLACEMENT COMMUNICATION & TRACKING SYSTEM

Comprehensive Technical Progress Report & Phase-Wise Implementation Roadmap

Project Version: v1.0 (Core Modules Complete)	**Tech Stack:** React (Vite) + Node/Express
Project Type: College Minor Project	**Report Date:** October 2026
User Roles: TPO, Recruiters, Students	**Automated Test Suite:** 21/21 Passing (100%)

1. Executive Summary & Problem Statement

Campus placement drives in colleges typically suffer from fragmented communication. Job openings, criteria, student resumes, application statuses, and interview outcomes are frequently scattered across email threads, messaging groups, and spreadsheets. This leads to missed deadlines, student confusion, duplicate applications, and heavy administrative burden on the Training and Placement Officer (TPO).

The **Campus Placement Communication & Tracking System** resolves this by establishing a centralized, role-based portal that connects all placement stakeholders: **(1) External Recruiters** who register, submit recruitment drives, define eligibility rules, and manage multi-round shortlists; **(2) TPO / Administrators** who govern recruiter access, review and approve drives, publish opportunities, view analytics, and audit logs; and **(3) Students** who maintain academic profiles, upload resumes, verify real-time eligibility, apply with one click, and track job offers.

Key Architectural Objectives & Principles

- **Zero Confusion:** Unified transparent portal for company schedules, eligibility criteria, and results.
- **Automated Governance:** Server-enforced eligibility prevents ineligible applications automatically.
- **Data Security & RBAC:** Strict role separation ensures recruiters cannot access unrelated students or drives.
- **Viva Readiness:** 1-Click quick login switcher bar and database reset mechanism for rapid examiner demonstration.

2. Phase-by-Phase Implementation Status Matrix

The project development was divided into systematic milestone phases from core architecture to automated testing:

Phase	Module Title	Key Deliverables Completed	Status
Phase 1	Core Architecture & RBAC Auth	JWT Authentication, bcrypt hashing, 3 User Roles (TPO, Company, Student), Quick Viva Demo Switcher bar, seed reset.	**COMPLETED**
Phase 2	Placement Drive Lifecycle	Recruiter drive submission, TPO review & approval/rejection with remarks, drive publication to student portal.	**COMPLETED**
Phase 3	Automated Eligibility Engine	Real-time eligibility evaluation on branch, CGPA, backlogs, batch year with specific failure breakdown banners.	**COMPLETED**
Phase 4	Application Pipeline & Offers	One-click application with cover note, duplicate check, recruiter multi-round candidate board, student offer accept/reject.	**COMPLETED**
Phase 5	Reports, Analytics & Audit	Placement summary KPIs (Placed %, Avg CTC), Branch analytics, CSV export, TPO system audit log trail, in-app notifications.	**COMPLETED**
Phase 6	Recruiter Verification & Resumes	Public recruiter signup with TPO verify/reject; student resume upload (PDF/DOC/DOCX, max 5MB), streaming view with token, disk cleanup.	**COMPLETED**
Phase 7	Quality Assurance & Testing	21 automated test cases passing in Node.js test runner covering auth, validation, streaming, role permissions, cleanup; Vite production build verified.	**COMPLETED**
Phase 8	Future Scope & Enhancements	Visual monthly recruitment calendar, student application withdrawal, simulated email dispatch, cloud MongoDB Atlas migration.	**PLANNED**

System User Roles & Access Boundaries

Role	Primary Permissions & Responsibilities	Access Scope
TPO / Admin	Verify recruiters, approve/reject drives, publish drives, export CSV, view audit logs, manage students	College-wide access
Company / Recruiter	Submit drives, configure criteria, review applicants, update interview rounds, publish final results	Own company drives only
Student	Update academic profile, upload resume (PDF/DOCX), verify drive eligibility, apply, respond to offers	Own applications & profile

3. Technical Implementation - Phases 1 to 4

Phase 1: Core Architecture, RBAC & Viva Demo Quick Switcher

Established the baseline MERN architecture with robust authentication and a Viva Demo Bar for effortless presentation.

- **JWT Authentication:** 7-day signed tokens containing user ID, role, and email.
- **Security Middleware:** requireAuth (extracts Bearer header and ?token= query params) and requireRole.
- **Quick Viva Switcher:** 1-Click login bar allows toggling between TPO, Google, Microsoft, TCS, and various student personas.
- **Database Seed Reset:** POST /api/auth/seed-demo reloads clean, consistent demo data anytime during viva testing.

Phase 2: Placement Drive Lifecycle & Workflow Management

Digitized the company recruitment visit workflow from initial draft to official campus publication.

- **Drive Creation:** Recruiter specifies CTC package, job role, locations, deadline, and eligibility criteria.
- **TPO Governance:** Drives enter pendingApproval status. TPO reviews and approves or rejects with formal feedback.
- **Publication:** Approved drives are published to the student portal, triggering in-app alerts to eligible students.

Phase 3: Automated Eligibility Verification Engine

Engineered a real-time rule evaluator that matches student records against company criteria dynamically.

- **4-Point Rule Check:** Evaluates Allowed Branches, Minimum CGPA, Maximum Active Backlogs, and Target Batch Year.
- **Visual Explanations:** Displays green 'Eligible' badge or red 'Ineligible' badge with specific reasons (e.g. *'Backlogs exceed maximum of 0'*).
- **Server-Side Enforcement:** Prevents API bypass by re-verifying eligibility before inserting application records.

Phase 4: Application Pipeline & Multi-Round Candidate Tracking

Provided recruiters with an interactive candidate review board and students with a stage-by-stage tracker.

- **Application Submission:** One-click application with cover notes and duplicate application prevention.
- **Recruiter Candidate Board:** Displays applicants with branch, CGPA, backlogs, technical skill tags, and resume links.
- **Stage Transitions:** Multi-round updates (Applied → Aptitude Test → Technical Round → Final Interview → Selected/Rejected).
- **Offer Management:** Selected students receive formal job offers in their dashboard and can Accept or Reject.

4. Technical Implementation - Phases 5 to 7

Phase 5: Reports, Placement Analytics & System Audit Trail

- **Placement KPIs:** Real-time statistics: Total Registered Students, Placement Percentage, Average Package, and Highest CTC.
- **CSV Report Export:** /api/reports/export-csv exports placement records with student roll numbers, companies, and packages.
- **System Audit Trail:** All administrative events are permanently logged with actor name, role, timestamp, and action details.
- **In-App Notification Center:** Unread badges and instant alerts for drive announcements and application round outcomes.

Phase 6: Recruiter Verification & Resume Document Management

- **Recruiter Signup & Verification:** External recruiters register with pendingVerification status. TPO reviews company website, credentials, and verifies or rejects with remarks.
- **Resume File Upload:** Drag-and-drop zone supporting PDF, DOC, and DOCX formats up to 5MB, utilizing Multer disk storage.
- **Token-Authenticated Streaming:** Direct inline streaming via /api/students/resume?token=... allows opening resumes in new browser tabs securely without cross-origin issues.
- **Orphan File Cleanup:** Uploading a replacement resume or deleting a resume immediately unlinks the old file from disk, preventing server storage leaks.
- **Cross-Role Visibility:** Verified recruiters and TPO officers can inspect student resumes directly from the candidate board and student registry.

Phase 7: Quality Assurance & Automated Test Suite Verification

Automated test suite implemented in server/tests/resume.test.js using Node.js native test runner (**21/21 Passing Tests**):

Test Suite Group	Scenarios Evaluated	Expected HTTP	Result
1. Authentication & RBAC	Unauthenticated upload/download/delete; Recruiter upload attempt blocked	401 / 403	**4/4 PASS**
2. File Validation	Missing file payload; Disallowed extension (.txt/.exe); >5MB size rejection	400 Bad Request	**3/3 PASS**
3. Upload & Profile Sync	Valid PDF upload; DB metadata update; Audit trail logged; Old file disk removal	200 OK	**4/4 PASS**
4. Inline Streaming	Bearer token stream; ?token= query auth; TPO & Recruiter access; Cross-student 403	200 OK / 403	**5/5 PASS**
5. Application Pipeline	Drive applicant list includes student resumeFile, originalName, studentProfileId	200 OK	**1/1 PASS**
6. Deletion & Cleanup	Student resume delete; DB fields set to null; RESUME_DELETED audit; 404 verification	200 OK / 404	**4/4 PASS**

Production Build: npm run build transforms 1916 client modules in 2.39s with zero bundle errors.

5. College Viva Demonstration Script (2-Minute Execution Flow)

To present this project effectively in front of external examiners, execute the following 7-step sequence using the top Quick Demo Switcher bar:

Step 1: Recruiter Drive Submission:

Switch to 'Google Recruiter' → Click 'Submit New Drive' → Enter role 'Software Engineer', CTC 18 LPA, CGPA 7.5, Backlogs 0 → Submit. Point out the drive enters 'Pending Approval'.

Step 2: TPO Review & Publication:

Switch to 'TPO Admin' → View 'Pending Company Drives' → Click 'Approve Drive' with remarks → Click 'Publish Drive'. Point out the published status and instant notification.

Step 3: Student Eligibility & Backlog Handling:

Switch to 'Rahul Verma' (Eligible) → Point out green 'Eligible' banner. Then switch to 'Priya Patel' (has 1 backlog) → Show red 'Ineligible' banner with specific reason: *'Backlogs exceed maximum of 0'*.

Step 4: Academic Profile & Resume Upload:

In Student Dashboard → Go to 'My Academic Profile' → Drag and drop a resume PDF → Show uploaded file card → Click 'View' to demonstrate inline streaming in a new browser tab.

Step 5: Student Application Submission:

Navigate to 'Explore Drives' as eligible student → Click 'Apply' → Submit cover note. Show immediate application confirmation and duplicate prevention.

Step 6: Recruiter Candidate Review & Shortlisting:

Switch back to 'Google Recruiter' → Go to 'Applicant Tracking' → Inspect candidate → Click 'Resume' to view the uploaded PDF → Advance candidate round to 'Technical Interview'.

Step 7: TPO Reports, CSV Export & Audit Trail:

Switch to 'TPO Admin' → Show Placement Analytics KPIs and Branch breakdown → Click 'Export CSV' to download placement records → Go to 'Audit Trail' to show all recorded events.

6. Phase 8 Roadmap & Future Scope

The following enhancements represent the planned Phase 8 milestone and serve as the project's official Future Scope:

- **Visual Recruitment Calendar:** Monthly calendar highlighting company visit dates, aptitude test schedules, and interview slots.
- **Student Application Withdrawal:** Option for students to withdraw submitted applications before the deadline with recruiter notice.
- **Email & SMS Gateway Dispatch:** Integration with SendGrid / Nodemailer to send real-time email reminders and offer letters.
- **Cloud MongoDB Atlas Cluster:** Toggle between the local zero-config JSON persistence engine and a production MongoDB Atlas cluster.

Document maintained & updated continuously as implementation milestones are completed. Generated for College Minor Project Viva & Evaluation.